جامعة تبوك
University of Tabuk

Early Detection and Prevention of Choking in Children

(KIDSAFE BREATH)

KidSafe®
— BREATH —

Introduction

Choking is a leading cause of injury and death in infants and toddlers due to their narrow airways and inability to express distress. Current health monitoring systems are either too generic or lack real-time alert features designed specifically for children. This creates a critical delay in responding to suffocation incidents.

Problem Overview

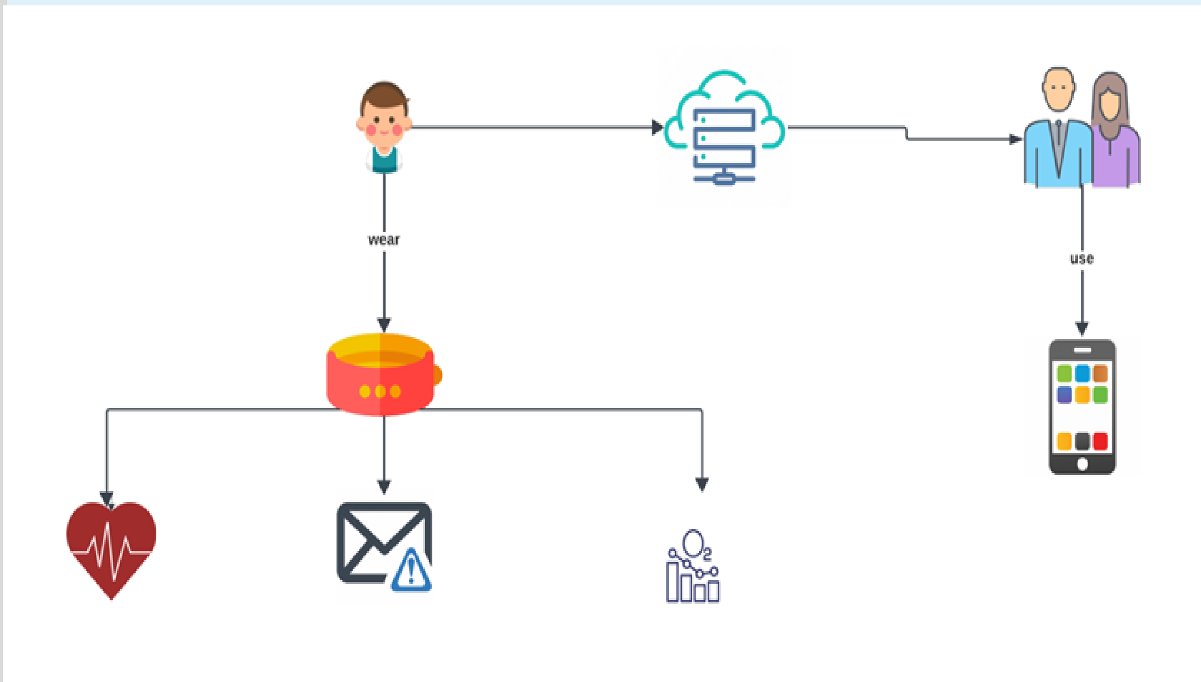
Choking is one of the leading causes of unintentional injury-related deaths among infants and toddlers. Due to their limited ability to express distress and their narrow airways, choking incidents can escalate rapidly and become fatal without timely detection. Existing health monitoring solutions are not specifically designed for young children and often lack real-time response capabilities. As a result, there is a clear need for a dedicated system that continuously monitors vital signs and instantly alerts caregivers when a potential suffocation risk is detected.

Aims and Objectives

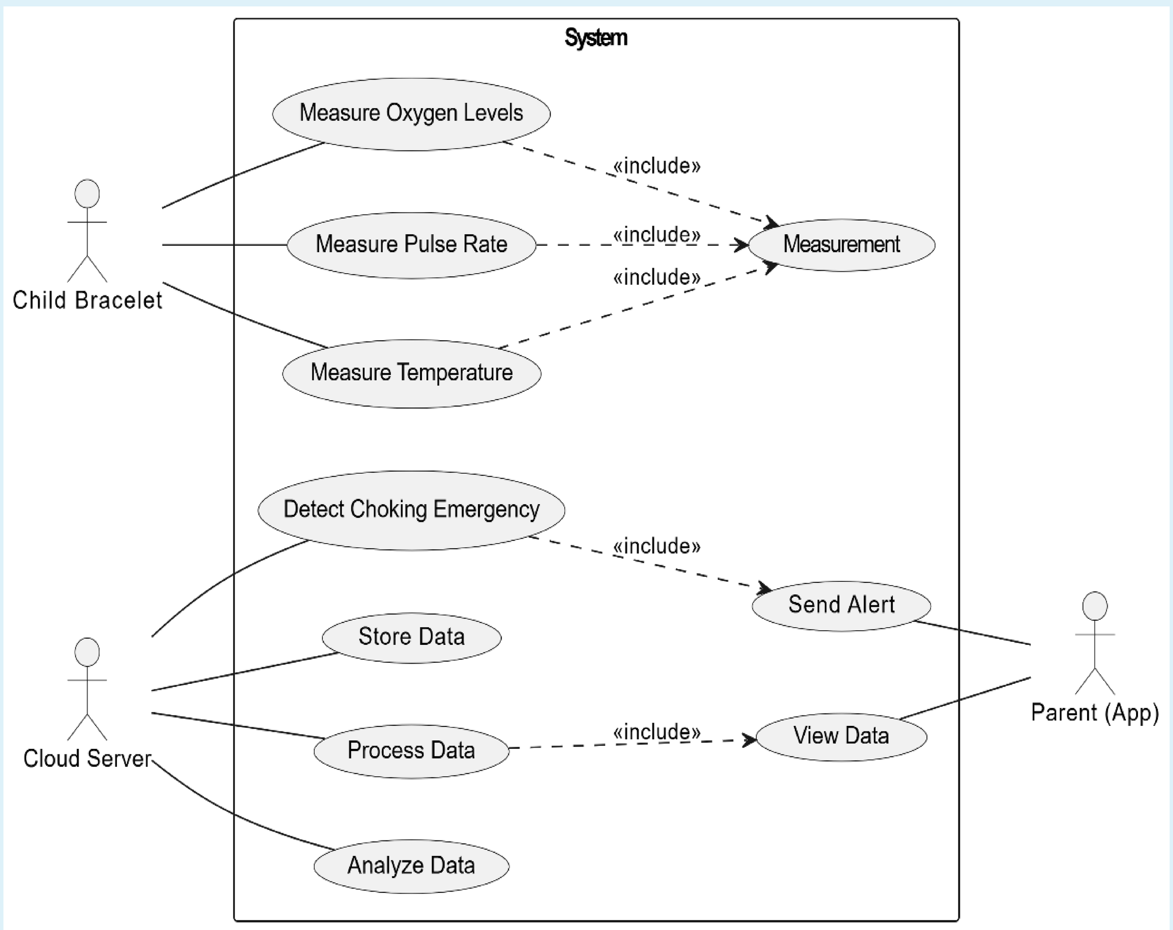
This project aims to design a smart bracelet that continuously monitors children's vital signs, including heart rate, oxygen saturation (SpO_2), and body temperature. It detects early signs of suffocation and sends real-time alerts to caregivers through a mobile app, using Firebase Realtime Database for secure data transmission. The goal is to improve emergency response time and enhance child safety through live health monitoring.

Methodology

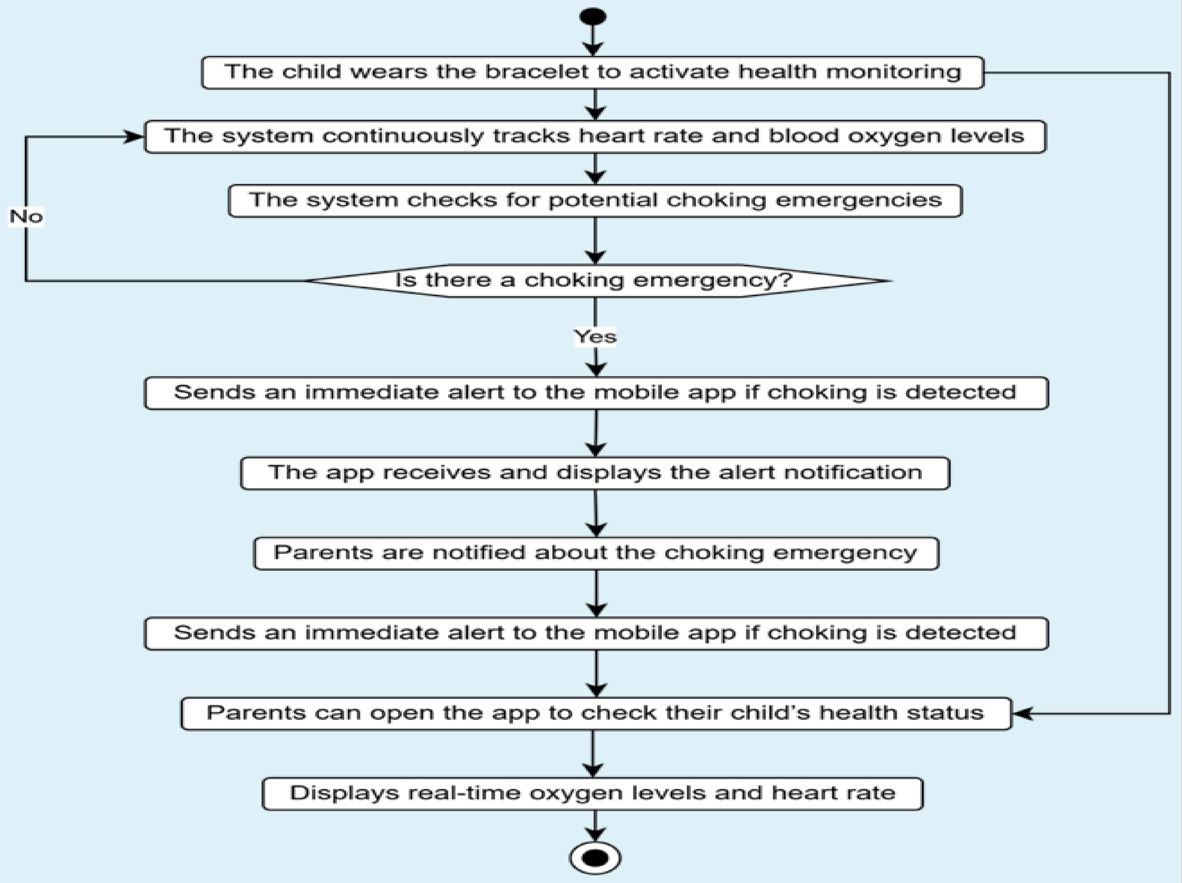
System Architecture



Use Case Diagram



Activity Diagram

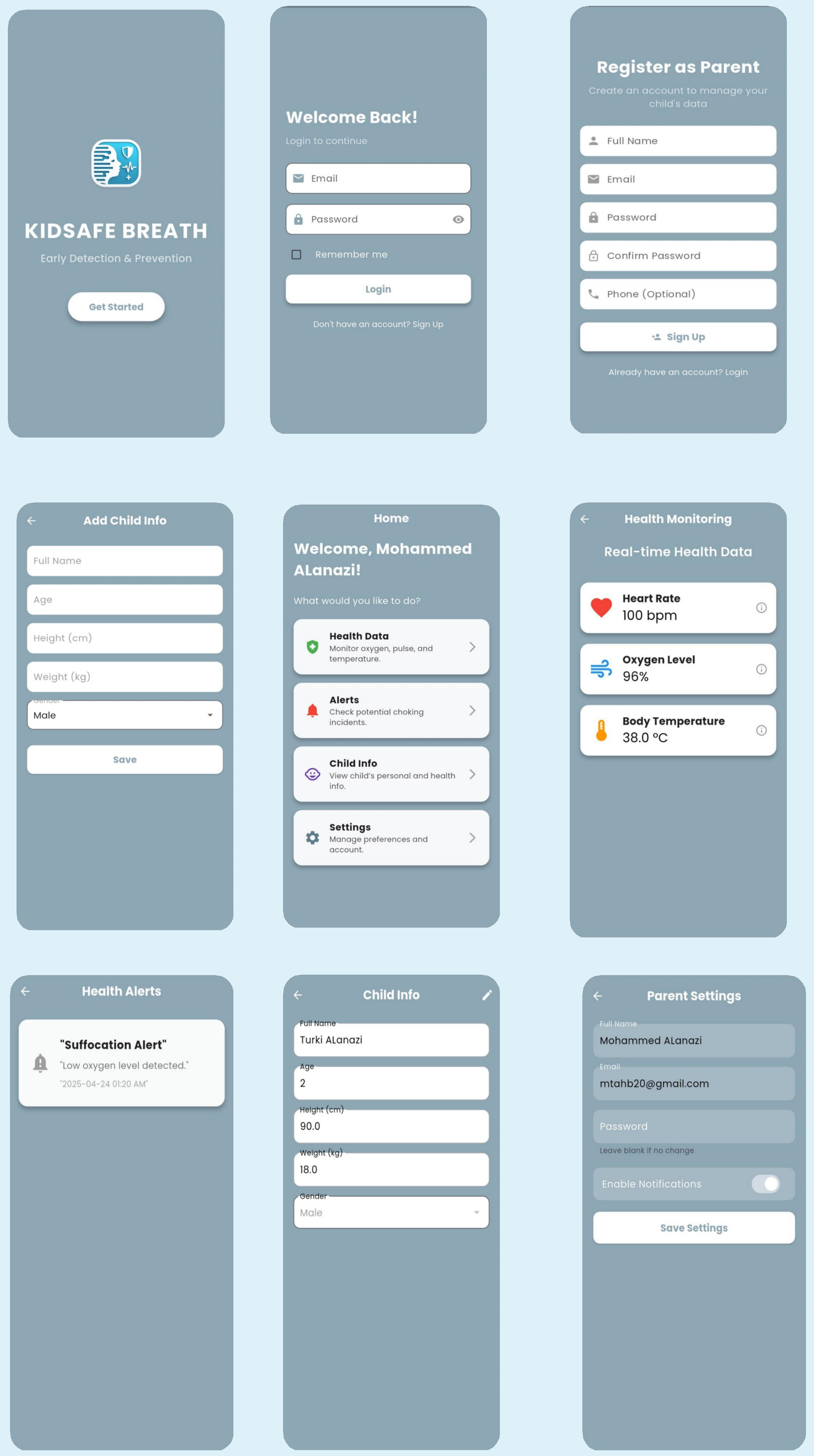


Results

The final output of the system includes a fully functional mobile app and a smart bracelet prototype. The images below show real-time data visualization, alert generation, and the final assembled hardware used to monitor vital signs in children.

Bracelet Design

UI Interfaces



Tools & Technologies

Flutter
Firebase
Visual Studio Code

Heart Rate and Oximeter (MAX30102)

Temperature Sensor (MLX90614-DCC)

ESP32 C6

Conclusion

KidSafe Breath demonstrates the successful integration of IoT and mobile technologies to support real-time health monitoring for children. The system effectively collects and transmits vital signs, detects abnormalities, and alerts caregivers instantly. This solution offers a practical and low-cost approach to early suffocation detection, enhancing child safety and enabling timely interventions.

Team Members

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